Problem-Driven Renewable Energy and Workplace Safety for Ghana: Techno-Economic Optimization of Rooftop Solar in Accra and Act 328 OHS Protocols

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ABSTRACT

Urban residential electricity consumers across Ghana experience severe utility tariff increases alongside recurring transmission outages, yet decentralized rooftop solar adoption remains impeded by financing hurdles and safety deficits. This research determines the top five energy and occupational safety obstacles in Ghana and engineers a techno-economic sizing engine paired with an Act 328 workplace safety protocol. Using national telemetry from the Energy Commission of Ghana, Public Utilities Regulatory Commission (PURC), and NASA POWER solar irradiance data (5.14 kWh/m2/day annual GHI), candidate problems were evaluated across formal multicriteria ranking metrics. This paper targets the five definitive energy and occupational safety bottlenecks in Ghana: grid distribution unreliability, high residential solar capital costs, amateur system sizing failures, toxic battery e-waste dumping, and installation fall and electrocution hazards. As a tangible open-source artifact, we developed and validated a rooftop solar PV sizing, LCOE, and PURC tariff payback engine with integrated Act 328 OHS protocols in /prototypes/energy/.

Keywords: Renewable Energy, Solar Photovoltaic, LCOE, PURC Electricity Tariffs, Accra, Occupational Health and Safety, Act 328

PROBLEM-TO-SOLUTION SUMMARY & OPEN ARTIFACT

This publication presents an empirical pathway connecting established computational and engineering theory to physical, field-validated artifacts addressing five ranked national problems in Ghana. A runnable, tested open-source prototype has been authored and published under an MIT license in the repository directory /prototypes/energy/ with automated unit test suites.

Canonical Web Reader: https://bernardfiagbenu.vercel.app/research/energy